PARISHRAM 2.0 2026 (Class 12th)

Mathematics

DPP:01

Relations and Functions

Q1 If A = {1, 2, 3} and R_1 = {(1, 2), (3, 2), (1, 3)}, R_2 = {(1, 3), (3, 6), (2, 1), (1, 2)} then:
(A) R_1 is a relation and R_2 is not a relation on A
(B) R_1 and R_2 both are relation on A
(C) R_1 and R_2 both are not relation on A
(D) None of these

Q2 Let R be a relation in the set N given by $R = \{(a, b) : a = b - 2, b > 6\}$. Then
(A) $(8, 7) \in R$ (B) $(6, 8) \in R$
(C) $(3, 8) \in R$ (D) $(2, 4) \in R$

Q3 If R = {(x, y) : x, y ∈W, 2x + y = 5} . Then
(A) $R = \{(\frac{5}{2}, 0), (2, 1), (\frac{3}{2}, 2), (\frac{1}{2}, 4), (0, 5)\}$
(B) $R = \{(\frac{5}{2}, 0), (2, 1), (\frac{3}{2}, 2), (\frac{1}{2}, 4)\}$
(C) $R = \{(0, 5), (1, 3), (2, 1)\}$
(D) $R = \{(1, 2), (3, 1), (5, 0)\}$

Q4 The relation R defined on the set of natural numbers as {(a, b) : a differs from b by 3), is given by
(A) {(1, 4), (2, 5), (3, 6) ,.....}
(B) {(4, 1), (5, 2), (6, 3) ,.....}
(C) {(1, 3), (2, 6), (3, 9) ,.....}
(D) None of the above

Q5 Let $n(A) = m$, and $n(B) = n$. Then, the total number of non-empty relations that can be defined from A to B is
(A) m^n
(B) $n^m - 1$
(C) $mn - 1$
(D) $2^{mn} - 1$

Q6 Two finite sets A and B have m and n elements respectively. If the total number of relation from A to B is 64, then the possible values of m and n can be
(A) 1 and 5 (B) 2 and 4
(C) 2 and 3 (D) 1 and 4

Answer Key

Q1	A
Q2	B
Q3	C
Q4	B
Q5	D
Q6	C

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

$A \times A =$
$\{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3),$
$(3,1),(3,2),(3,3)\}$

$\because R_1 \subset A \times A \therefore R_1$ is a relation on A.

As (3, 6) $\in R_2$ but $6 \notin A \therefore R_2 \not\subset A \times A$

$\therefore R_2$ is a relation on A.

Video Solution:

Q2 Text Solution:

Given, R = {(a, b) : a = b – 2, b > 6}

Since, b > 6, so (2, 4) $\notin$ R

Also, (3, 8) $\notin$ R as 3 ≠ 8 – 2

and (8, 7) $\notin$ R as 8 ≠ 7 – 2

Now, for (6, 8), we have

8 > 6 and 6 = 8 – 2, which is true

$\therefore$ (6,8) $\in$ R

Video Solution:

Q3 Text Solution:

For x = 0, 2 × 0 + y = 5 $\Rightarrow$ y = 5 $\in$ W

For x = 1, 2 × 1 + y = 5 $\Rightarrow$ y = 3 $\in$ W

For x = 2, 2 × 2 + y = 5 $\Rightarrow$ y = 1 $\in$ W

For x = 3, 3 × 2 + y = 5 $\Rightarrow$ y = –1 $\notin$ W

We stop have as for x > 3, y is not a whole number

$\therefore R = \{(0, 5), (1, 3), (2, 1)\}$

Video Solution:

Q4 Text Solution:

Given, R = {(a, b) : a – b = 3} = {(4,1), (5, 2), (6, 3),...}

Video Solution:

Q5 Text Solution:

Given, $n(A) = m$ and $n(B) = n$

$\therefore$ Total number of relations from A to $B = 2^{mn}$

$\therefore$ Total number of non-empty relations from A to $B = 2^{mn} - 1$

Video Solution:

Q6 Text Solution:

Clearly, n(A × B) = n(A) × n(B) = mn

$\therefore$ Total number of relation from A to B = 2^{mn}

Thus, we have $2^{mn} = 64 = 2^6$

$\Rightarrow$ mn = 6

Hence, possible value of m and n are 2 and 3, respectively.

Video Solution:





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